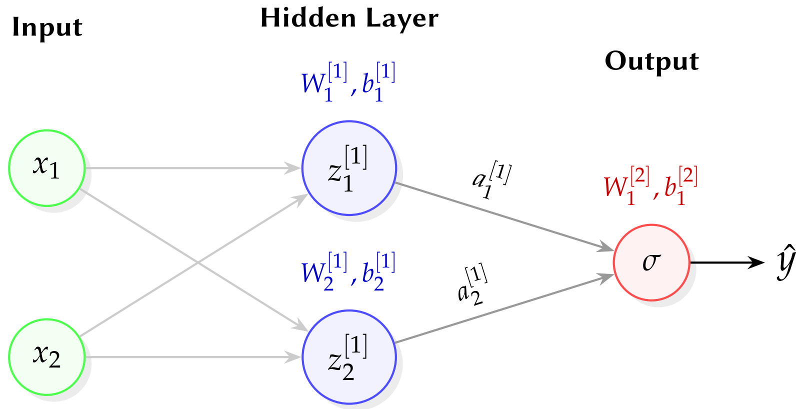
Classwork 02//Training a Neural Network

Student Name: ______________________ **Points:** 30

A Number: ______________________ **Due:** Feb 12, 2026

1 Neural Network

Consider a Neural Network as follows: **(10 Points)**



We consider two features as input, with each neuron having Leaky ReLU as activation layer with negative slope coefficient $\alpha = 0.01$.
Consider the initial weights as

1. $W_1^{[1]} = [0.1, -0.2]$, and $b_1^{[1]} = 0.1$
2. $W_2^{[1]} = [0.2, 0.1]$, and $b_2^{[1]} = 0.2$
3. $W_1^{[2]} = [-0.3, 0.1]$, and $b_1^{[2jj]} = 0.1$.

Given the input sample as $\mathbf{x} = \{0.2, 0.1\}$ and known response variable as $y = x_1 + x_2 = 0.3$ what is the predicted $\hat{y}$. What is the squared error from the resulting network operation? Show your work.

1.1 Solution

$$z_1^{[1]} = 0.2 \times 0.1 - 0.2 \times 0.1 + 0.1 = 0.1$$
$$z_2^{[1]} = 0.2 \times 0.2 + 0.1 \times 0.1 + 0.2 = 0.25$$

Since we are using LeakyRELU, and $z_1^{[1]} > 0$, $z_2[1] > 0$, $a_1^{[1]} = 0.1$, $a_2[1] = 0.25$.

$$z_1^{[2]} = 0.1 \times -0.3 + 0.25 \times 0.1 + 0.1 = 0.095 = a_1^{[2]} = \hat{y}$$

Squared error: $(0.3 - 0.095)^2 = 0.042025$.

2 Early Stopping

We train a neural network with early stopping (patience = 2 which means stop if validation loss does not improve for 2 consecutive epochs). The loss values across 8 epochs are: **(10 Points)**

Table 1: Neural Network Training and Validation Loss per Epoch

Epoch	Training Loss	Validation Loss
1	2.5	2.4
2	2.2	2.1
3	1.9	1.8
4	1.7	1.7
5	1.5	1.8
6	1.3	1.9
7	1.1	1.9
8	1.0	2.0

1. At which epoch do we trigger early stopping?
2. Which epoch's model do we keep (the **best** model)?

2.1 Solution

1. After epoch 6, we trigger early stopping

Epoch	Val Loss	Best So Far	Improved?	Patience Counter	Action
1	2.4	2.4	Yes (init)	0	Continue
2	2.1	2.1	Yes	0	Continue
3	1.8	1.8	Yes	0	Continue
4	1.7	1.7	Yes	0	Continue
5	1.8	1.7	**No**	1	Continue
6	1.9	1.7	**No**	2	**STOP**

2. From the previous table, it is clearer the model trained after the completion of Epoch 4 is the best model that we should keep.

3 Learning Rate Scheduler

We use a step decay scheduler with: **(10 Points)**

1. Initial learning rate $\eta_0 = 0.1$
2. Decay rate $\gamma = 0.5$
3. Step size $s = 3$, i.e. learning rate decays after every three steps.

Based on the above description,

1. write down the formula for step decay learning rate $\eta(t)$ as a function of epoch t and should be parametrized by η_0, γ and s.
2. Calculate $\eta(t)$ for epoch $t = 4$, $t = 7$, and $t = 9$.

3.1 Solution

1. $$\eta(t) = \eta_0 \gamma^{\lfloor \frac{t}{s} \rfloor}$$

   where $\lfloor\rfloor$ is the floor function.

2. Putting the values in the formula

$$\begin{aligned}
\eta(t) &= \eta_0 \gamma^{\lfloor \frac{t}{s} \rfloor} \\
\eta(4) &= 0.1 \times 0.5^{\lfloor \frac{4}{3} \rfloor} = 0.05 \\
\eta(7) &= 0.1 \times 0.5^{\lfloor \frac{7}{3} \rfloor} = 0.025 \\
\eta(9) &= 0.1 \times 0.5^{\lfloor \frac{9}{3} \rfloor} = 0.0125
\end{aligned}$$